

Oil India Limited — HSSE SIF Platform Implementation Report

Enterprise AI-driven Incident Classification, Precursor Risk Analytics, Safety Knowledge Graph, and Compliance Enforcement System
Aligned with OISD-STD-105, DGMS OMR 2017, and IOGP Life-Saving Rules

Target Organization	Oil India Limited (HSSE Department)
Project Status	Fully Implemented & Verified Baseline Platform
Operational Coverage	12+ Oil India Installations (Baghjan, Duliajan, Digboi, Moran)

1. Executive Overview

In upstream oil and gas operations—spanning drilling rigs, gas gathering stations, refineries, and high-pressure cross-country pipelines—traditional safety tracking often struggles to separate high-consequence Serious Injury & Fatality (SIF) precursors from routine low-severity observations.

This platform provides an end-to-end artificial intelligence and data-driven triage solution developed for Oil India Limited (HSSE Department). Built with a zero-external-framework Python server and offline-first ML models, it automatically ingests field safety reports (in English, Hindi, Hinglish, or Assamese regional terms), evaluates energy pathway hazards, checks safety barrier health, flags IOGP Life-Saving Rule violations, builds interactive safety knowledge graphs, and forecasts temporal risk trends across 12+ Oil India operational installations.

2. Summary of Implemented Features & Core Novelties

Feature 1: AI Incident Classifier & SIF Precursor Triage

- **Domain NLP Tokenizer:** Ingests technical oilfield jargon (BOP, flare stack, wellhead pressure, H2S sensors, wireline) and normalizes mixed Hinglish/Assamese field inputs.
- **Dual ML Engine:** Binary classification model for SIF precursor detection and multi-class model predicting 10 IOGP Life-Saving Rules.
- **Numeric SIF Scoring:** Outputs a 0–100% SIF probability index and automated audit rationale with OISD recommendations.

Feature 2: Safety Relationship Map (D3.js Knowledge Graph)

- **Graph Extraction API:** Dynamically extracts nodes and edges linking Activity -> Hazard -> Barrier -> IOGP Rule -> Consequence.
- **Interactive D3.js Renderer:** Force-directed topology map with node filtering, search, physics freeze controls, and node-dragging mechanics.

Feature 3: Historical Similar Incident Retrieval Engine

- **Cosine Similarity Search:** Pre-computes TF-IDF vector matrix over 500+ historical safety observations.
- **Near-Miss Retrieval:** Instant search returning top matching incidents, past barrier failures, and previously applied corrective measures.

Feature 4: Temporal Risk Forecasting & KPI Dashboard

- **Polyfit Trend Extrapolation:** Polynomial regression calculating 30-day and 60-day projected SIF precursor rates.

- **Real-Time Visualizations:** Interactive charts for Monthly Trends, Hazard Breakdown, Barrier Conditions, and Severity Donut Chart.

Feature 5: Hierarchy of Controls Remediation Engine

- **5-Tier Categorization:** Automatically maps incident risks into Elimination, Substitution, Engineering Controls, Administrative Controls, and PPE.

Feature 6: Multilingual & Hinglish Text Processing

- **Mixed Language Processing:** Handles mixed field text (English, Hindi, Hinglish, Assamese terms), converting colloquial technical entries into normalized tokens.

Feature 7: Role-Based Access Control (RBAC) System

- **Enterprise Roles:** Configured for HSE Manager, Site Manager, Field Supervisor, and Safety Analyst with scoped permissions.

Feature 8: Multi-Site Hero Carousel & Visual Monitoring

- **Auto-Rotating Hero Slideshow:** 2.5-second rotation speed with smooth 0.45s fade transitions across Baghjan Field #5, Duliajan GGS, Digboi Refinery, and Moran OCS Station.

Feature 9: UTF-8 Excel-Compatible Data Export

- **One-Click CSV Export:** Serves safety report records with embedded UTF-8 Byte Order Mark (BOM), enabling Microsoft Excel to natively open columns cleanly.

3. REST API Specifications

Method	Endpoint	Description
GET	/	Serves main Single-Page Application (public/index.html)
POST	/api/classify	Runs ML inference for SIF risk, IOGP rule & corrective actions
POST	/api/similar	Executes TF-IDF cosine similarity search for near-miss retrieval
GET	/api/analytics	Serves KPI summaries, hazard distributions & polyfit risk forecast
GET	/api/knowledge-graph	Serves D3 graph nodes & link relationship edges
GET	/api/reports	Serves safety observation records
GET	/api/export	Serves UTF-8 BOM encoded CSV download for Microsoft Excel